7th, KM Haridwar, National Highway Vardhmanpuram,
Roorkee, Rehmadpur, Uttarakhand 247667

Lab Sheet-02

Python Programming Lab – Control Statements and Loops

MCA I Semester (Session 2026-2027)

Objectives:

- To implement Python programs using conditional logic for solving decision-based problems.
- To use iterative statements (for, while, nested loops) for repetitive computational tasks.

Requirements:

- Python 3.x installed on the system (IDLE, PyCharm, VS Code, or any preferred IDE).
- Basic understanding of **if**, **if-else**, **elif** statements, and looping constructs.
- Familiarity with number operations, string handling, and mathematical concepts like prime numbers, factorial, etc.

Program List:

1. Check if a number is a multiple of both 3 and 7.
2. Check if a number is a palindrome.
3. Check if a number is an Armstrong number.
4. Display a grade based on marks:
   - A: 90–100
   - B: 75–89

- C: 50–74
- F: below 50

5. Find the absolute value of a number.
6. Check whether a triangle is valid given its angles.
7. Determine the type of a triangle (equilateral, isosceles, scalene).
8. Check if a string contains a particular substring.
9. Check eligibility for voting.
10. Find the smallest among three numbers.
11. Print numbers from 1 to 10 using a for loop.
12. Print numbers from 10 to 1 using a while loop.
13. Print the sum of first n natural numbers.
14. Print the multiplication table of a given number.
15. Print all even numbers between 1 and 50.
16. Print all odd numbers between 1 and 50.
17. Find the factorial of a given number.
18. Print Fibonacci series up to n terms.
19. Reverse a number using a loop.
20. Count the number of digits in a number.
21. Find the sum of digits of a number.
22. Print all prime numbers between 1 and 100.
23. Check if a number is perfect.
24. Print a Pattern:

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```
*
* *
* * *
* * * *
* * * * *
```

25. Print a reverse Pattern:

```
* * * * *
* * * *
* * *
* *
*
```

26. Print a pyramid pattern of numbers.
27. Display the multiplication table from 1 to 10.
28. Find the HCF of two numbers using a loop.
29. Find the LCM of two numbers using a loop.
30. Check if a number is strong (sum of factorial of digits equals the number).

Instruction:

- Each program should be written in a separate .py file named according to the question number (e.g., Q1_multiple_3_7.py).
- Use user input for all required values.
- Add comments explaining the logic used.
- Output should be well formatted for clarity.
- Test the program with multiple test cases.

Guideline:

- Maintain proper indentation (4 spaces) and coding style.

- Avoid using pre-built library functions unless explicitly allowed.
- Focus on logical accuracy before optimizing the code.
- Ensure there are no syntax or runtime errors before submission.
- Use descriptive variable names for better readability.

Submission:

- Submit the source code files (.py) for each task along with a brief report documenting the implementation details and the results of the executions.



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